onero Robotic Arm A1 User Guide

Preliminary Notes

The following applies throughout this document and will not be restated in subsequent sections.

- This robotic arm is sold under the product name onero Robotic Arm A1, hereinafter referred to as "A1 Robotic Arm" or "A1".
- The A1 Robotic Arm is a product of OneRobotics (Shenzhen) Co., Ltd., hereinafter referred to as "onero".

1. Quick Start Guide

1.1 Important Safety Information

This chapter contains important safety information. All individuals and organizations must read and understand this information before using the device, especially before powering it on for the first time. It is essential to follow and comply with all assembly instructions and guidelines in this manual. Pay particular attention to text with warning symbols. Before operating the device, be sure to obtain and read the robotic arm user manual. If you have any questions about operation, please contact us.

How to reach us:

- Email: support@switch-bot.com
- Phone:
  - China: 400-822-5888 (UTC+8 10:00-12:20, 14:00-22:00, except public holidays)
  - North America: +1 866-530-2887 (UTC-4 10:00-19:00, except public holidays)
  - Japan: 0120-676-162 (UTC+9 11:00-13:20, 14:40-23:00, except public holidays)
- Other regions: Please contact us via email or through the SwitchBot official app.

1.1.1 Warning Icon

⚠ This symbol indicates a potentially hazardous situation that, if not avoided, may result in personal injury, property damage, or serious equipment damage.

1.1.2 Disclaimer

onero assumes no responsibility for any damage resulting from any user-induced damage, alteration, or modification to the robotic arm. onero assumes no responsibility for any damage to the robotic arm or any other equipment resulting from programming errors, operational errors, or equipment malfunctions.

1.1.3 Limitation of Liability

By using the robotic arm, you are deemed to have read, understood, acknowledged, and accepted all terms and contents of this product manual and safety information. The user agrees to take full responsibility for their own actions and all resulting consequences. The user agrees to use the robotic arm only for lawful purposes and to accept these terms as well as any related policies or guidelines that onero may establish. During operation, strictly adhere to and follow the requirements outlined in the user manual and safety information (but not limited to the foregoing). onero assumes no liability for any personal injury, accidents, property damage, legal disputes, or other disputes arising from improper use or force majeure.

The A1 Robotic Arm is not intended for use by individuals under the age of 18 or those without full legal capacity. Please ensure that such individuals do not come into contact with this product, and take extra precautions when operating the device in their presence.

The information contained in this manual does not encompass the complete design, installation, and operation of robotic arm applications (certain installation tutorials and drag teaching demonstration videos are included), nor does it address all peripheral devices that may affect the safety of the overall system. The design and use of the complete system must comply with the safety requirements prescribed by the standards and regulations applicable in the country or region where the robotic arm is installed.

Robotic arm integrators and end customers are solely responsible for ensuring compliance with all relevant regulations and applicable laws, and for ensuring that no significant hazards exist within the entire robotic arm application. Such responsibility includes, but is not limited to, the following:

1.1.3.1 Authority and Responsibility

- Conduct a comprehensive risk assessment of the entire robotic arm system.
- Connect additional safety devices for other machinery as determined by the risk assessment.
- Ensure the accurate design and installation of the entire robotic arm system, including both software and hardware components.
- Integrators and end customers shall comply with all relevant regulations and applicable legal requirements when conducting safety assessments, and shall ensure that the robotic arm poses no significant hazards or safety risks in actual use.

- Identify and evaluate all potential safety risks prior to operating the device.
- Ensure that no safety measures are altered or circumvented by any user.
- Compile and maintain all documents and records constituting the technical documentation, including risk assessments and this manual.

1.1.3.2 Environment

- Before first use, read this manual carefully to understand the basic operating procedures and usage guidelines.
- Select a sufficiently open workspace for operation. The robotic arm is not equipped with automatic obstacle avoidance or proximity sensing capabilities.
- Operate the robotic arm only within an ambient temperature range of 5 °C to 50 °C.
- Mount the robotic arm on a table surface or install it on a designated base. The mounting surface must be level, stable, and free from vibration.

1.1.3.3 Inspection

- Verify that the robotic arm shows no visible abnormalities prior to use.
- Ensure that all cable connections are correctly secured during operation.

1.1.3.4 Operation

- Ensure that the surrounding area is sufficiently clear during operation.
- Operate the robotic arm within direct line of sight at all times.
- The maximum payload of the robotic arm is 3 kg. Ensure the load does not exceed this limit during operation.
- If the device exhibits any abnormal behavior, cease operation immediately to prevent secondary injury.
- In the event of any abnormality, contact qualified technical personnel. Do not attempt to resolve the issue independently.

1.1.3.5 Warnings

- Ensure the robotic arm is at all times correctly and securely fixed in place.
- If entry into the robotic arm's workspace is required, do so only after taking all necessary personal safety precautions. Wear safety goggles and appropriate protective equipment where necessary.
- Ensure the robotic arm has sufficient clearance to operate freely.
- Ensure that all safety measures are implemented as determined by the risk assessment.

- Do not wear loose clothing when operating the robotic arm. Secure long hair prior to operation.
- Do not operate the robotic arm if it is damaged or exhibits any abnormality.
- Keep all body parts, including the head and face, clear of the robotic arm and any area within its operational reach during operation.
- Do not modify the robotic arm in any manner. Unauthorized modifications may introduce unforeseen hazards and risks to the operator, the equipment manufacturer, and integrators.
- Do not expose the robotic arm to strong or permanent magnetic fields, as these may cause damage to the arm.
- The robotic arm generates heat during operation. Do not touch the robotic arm while it is in operation or immediately after it has stopped, as prolonged contact may cause discomfort or injury. Disconnect the power supply and allow at least one hour for the arm to cool before handling.
- The A1 Robotic Arm is not intended for use by individuals under the age of 18 or those without full legal capacity.
- **This product is not equipped with a holding brake.** Do not subject the robotic arm to sustained heavy loads at maximum torque output for extended periods, as this may cause motor failure due to overheating, resulting in an uncontrolled free drop of the arm and potentially causing property damage or personal injury. Should the robotic arm drop suddenly as a result of the user's failure to observe this warning — including but not limited to prolonged operation under heavy load or use beyond the product's rated specifications — the company shall bear no legal liability and shall not be obligated to provide any compensation for any resulting personal injury (including but not limited to crush injuries, impact injuries, or entrapment) or property damage.
- **This product must be installed strictly in accordance with the installation tutorial videos provided by onero** (see the GitHub links in Section 2.4 — Basic Installation Guide). Failure to do so may result in zero-force drag teaching failure, accuracy drift, or other abnormal behavior. If you have any questions regarding the installation process, contact us prior to proceeding. Do not attempt alternative installation methods independently. Unauthorized changes to the installation method may necessitate robotic arm parameter recalibration or introduce unforeseen risks. Should the robotic arm sustain damage, experience functional failure, or cause any personal injury or property loss as a result of the user's failure to follow the official installation tutorial, the company shall bear no liability whatsoever.

2. Product Overview

This 7-DOF (7 degrees of freedom) robotic arm is designed for education and research, consumer applications, and industrial automation. With a maximum payload of 3 kg, its 7 rotary joints provide comprehensive operational flexibility while ensuring high precision and repeatability. The arm features a modular design for easy maintenance and upgrades. It offers extensive open-source resources that simplify programming and operation, enabling even non-professionals to get up and running quickly. It can be widely applied in scientific research, education, automotive manufacturing, electronics assembly, food processing, laboratory automation, and home services.

2.1 Package Contents

This robotic arm product is named: onero Robotic Arm A1. The suffixes L, R, and S correspond to left arm, right arm, and dual arm respectively.

Product	onero Robotic Arm A1L	onero Robotic Arm A1R	onero Robotic Arm A1S
Items Included	A1 Left Arm Unit × 1	A1 Right Arm Unit × 1	A1 Dual Arm Unit × 1
	USB-to-CAN adapter (for CAN debugging) × 1	USB-to-CAN adapter (for CAN debugging) × 1	USB-to-CAN adapter (for CAN debugging) × 2
	Micro USB Cable × 1	Micro USB Cable × 1	Micro USB Cable × 2
	Power & CAN Communication Cable × 1	Power & CAN Communication Cable × 1	Power & CAN Communication Cable × 2
	Mounting Screws × 5	Mounting Screws × 5	Mounting Screws × 10
	Hex Key × 1	Hex Key × 1	Hex Key × 1

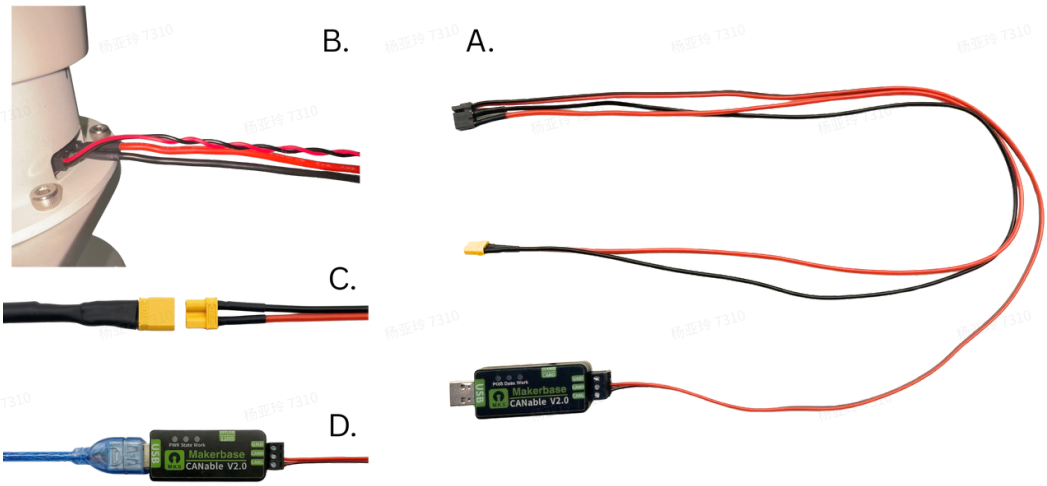
2.2 Basic Usage

The robotic arm has 7 degrees of freedom and a maximum payload of 3.0 kg. Its 7 rotary joints provide comprehensive operational flexibility, ensuring high precision and repeatability. The

lightweight design enables fast motion while maintaining a high payload capacity. It can be widely used in embodied intelligence applications for real-world data collection.

2.2.1 CAN Setup and Connection

2.2.1.1 CAN Setup



- A. Power and CAN communication cable
- B. Motor CAN port connection
- C. Power adapter connection
- D. CAN-to-USB adapter and USB extension cable connection

2.2.1.2 CAN Connection

Steps:

1. Route the CAN cable and connect the CANH and CANL wires to the CAN-to-USB adapter. As shown in D in the preceding figure, connect the red wire to CANH and the black wire to CANL.
2. Connect the CAN cable to the motor CAN port located below the first joint of the robotic arm, as shown in B in the preceding figure.

3. Connect the power cable to the power adapter, as shown in C in the preceding figure.

2.3 Drag Teaching / Demonstration Mode

Drag teaching mode is the core function of the robotic arm for task recording and playback. In this mode, the operator manually guides the arm through a series of target motions, and the system automatically records the movement trajectory. Once teaching is complete, the arm can automatically replay the recorded path.

This feature significantly reduces reliance on programming skills, enabling non-professionals to quickly complete task deployment and commissioning.

2.3.1 Safety Precautions

- Before entering drag teaching mode, confirm that no personnel or obstacles are present within the robotic arm's working range.
- During drag teaching, move the robotic arm slowly and at low speed. Avoid large or rapid dragging motions to reduce the risk of collisions or impacts due to misoperation.
- When using drag teaching (zero-force drag), zero-force control mode must be enabled. Never forcibly drag the robotic arm when it is unpowered or when the brake has not been released, as this may damage the reduction mechanism.

2.3.2 Pre-Teaching Preparation

- **Robotic Arm Status Check:** Confirm that the device is correctly installed and the power connection is normal.
- **Zero Position Confirmation:** Before each use or when switching operating modes, the robotic arm must first be moved to the zero position.

> ⚠ **Warning:** Zero position confirmation is required not only prior to drag teaching, but before every use and every mode switch. onero assumes no liability for any accuracy drift, arm malfunction, structural damage, personal injury, or property loss arising from the user's failure to confirm the zero position before operation. The user shall bear full responsibility for all resulting consequences.

2.3.3 Drag Teaching Operation

The following commands enable drag teaching functionality for zero-force drag recording and trajectory playback.

1. **Launch the robotic arm low-level driver node:**

Before performing any teaching operation, launch the robotic arm's low-level driver node `onero_driver`:

```
代码块
ros2 launch onero_driver <arm_type>_driver.launch.py
```

Enter <arm_type> corresponding to your actual arm model. Currently supported model identifiers are: `a1_l` (left arm) and `a1_r` (right arm).

2. **Launch the drag teaching node:**

```
代码块
ros2 run onero_examples drag_teaching_node
```

Once executed successfully, the system will launch the drag teaching node and enter the command-awaiting state.

3. **Feature Description**

- **Zero-Force Drag Recording:** After starting recording mode, the robotic arm enters zero-force control state. The operator can manually drag the arm through the target motions; the system automatically collects and records joint position, velocity, and torque data, then saves it to a file.
- **Trajectory Playback:** The system reads trajectory data from the recorded file and executes the following sequence: the arm automatically returns to the zero position → moves to the trajectory start point → plays back the complete trajectory in recorded order.

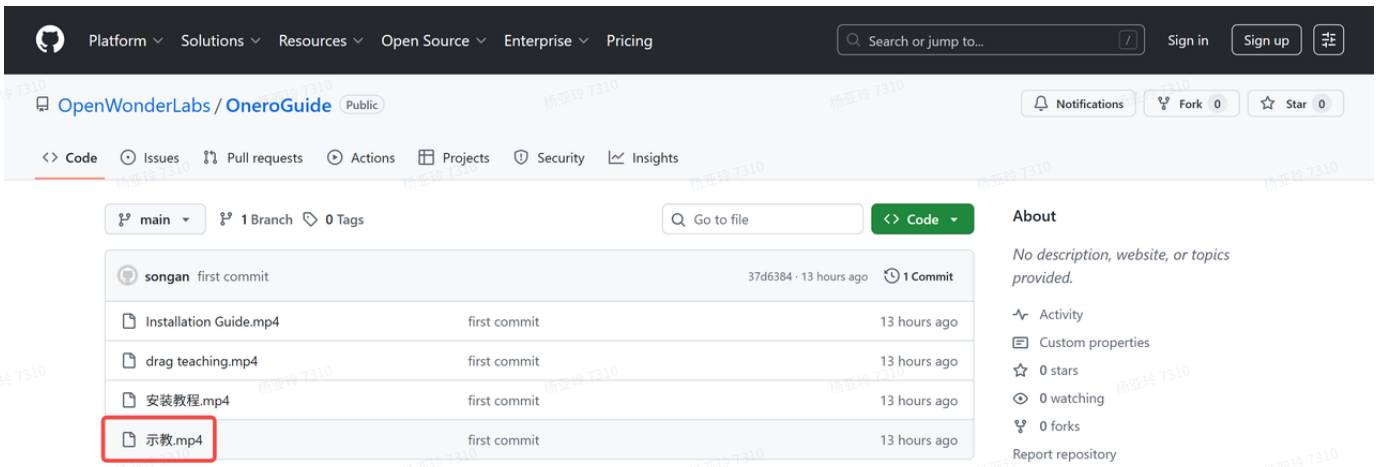
4. **Operation Guide**

- Enter `1` to activate zero-force drag mode.
- Manually guide the robotic arm through the desired teaching motions.
- Enter `2` to stop recording and save the trajectory.
- Enter `3` to play back the trajectory just recorded.
- Enter `4` to load and play back a previously saved trajectory.

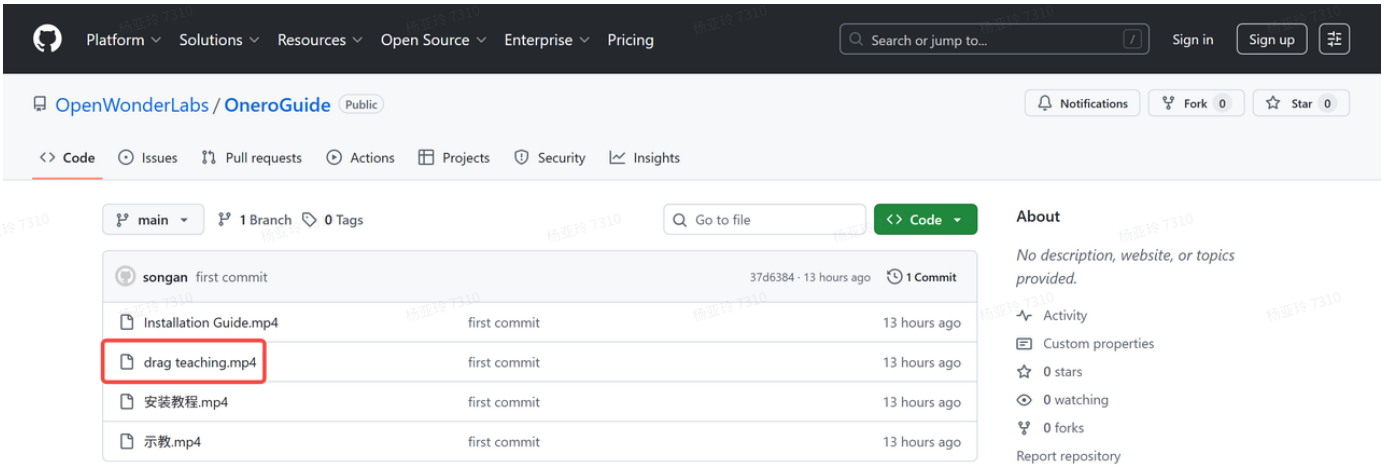
```
woan@woan-Y9000P-IAX10:~/test/WoanArm_for_develop$ ros2 run onero_examples drag_teaching_node
[INFO] [1774951973.080577881] [drag_teaching_node]: ========================================
[INFO] [1774951973.080651604] [drag_teaching_node]:   Drag Teaching Node Starting
[INFO] [1774951973.080658899] [drag_teaching_node]: ========================================
✓ DragTeaching initialized
✓ Serial port opened: /dev/ttyACM0
✓ Motor controller initialized (7 motors)
✓ DragTeaching mount orientation: vertical
✓ DragTeaching gravity vector: [9.81, 0, 0]
[INFO] [1774951985.606076601] [drag_teaching_node]: ✓ Hardware resources initialized successfully
[INFO] [1774951985.606130241] [drag_teaching_node]:   Device: /dev/ttyACM0
[INFO] [1774951985.606172938] [drag_teaching_node]:   Robot model: a1_l
[INFO] [1774951985.606204392] [drag_teaching_node]:   Mount orientation: vertical
[INFO] [1774951985.608323729] [drag_teaching_node]: ========================================
[INFO] [1774951985.608360790] [drag_teaching_node]:   Drag Teaching Node Ready
[INFO] [1774951985.608371023] [drag_teaching_node]: ========================================
[INFO] [1774951985.608380246] [drag_teaching_node]:   This node provides robot arm drag teaching:
[INFO] [1774951985.608388439] [drag_teaching_node]:   1. Zero-force drag recording: manually drag the arm, auto-record trajectory
[INFO] [1774951985.608398463] [drag_teaching_node]:   2. Trajectory replay: replay currently recorded trajectory
[INFO] [1774951985.608406820] [drag_teaching_node]:   3. Select trajectory replay: choose and replay from history
[INFO] [1774951985.608415295] [drag_teaching_node]: ========================================
[INFO] [1774951985.608423314] [drag_teaching_node]: >>> Enter command:
[INFO] [1774951985.608432268] [drag_teaching_node]:     0 - Stop all operations
[INFO] [1774951985.608440869] [drag_teaching_node]:     1 - Start zero-force drag recording
[INFO] [1774951985.608448947] [drag_teaching_node]:     2 - Stop recording
[INFO] [1774951985.608457762] [drag_teaching_node]:     3 - Replay current trajectory
[INFO] [1774951985.608465471] [drag_teaching_node]:     4 - Select history trajectory
[INFO] [1774951985.608473188] [drag_teaching_node]:     5 - Exit program
[INFO] [1774951985.608481354] [drag_teaching_node]: >>>
```

5. Video Demonstrations:

- Chinese version: https://github.com/OpenWonderLabs/OneroGuide



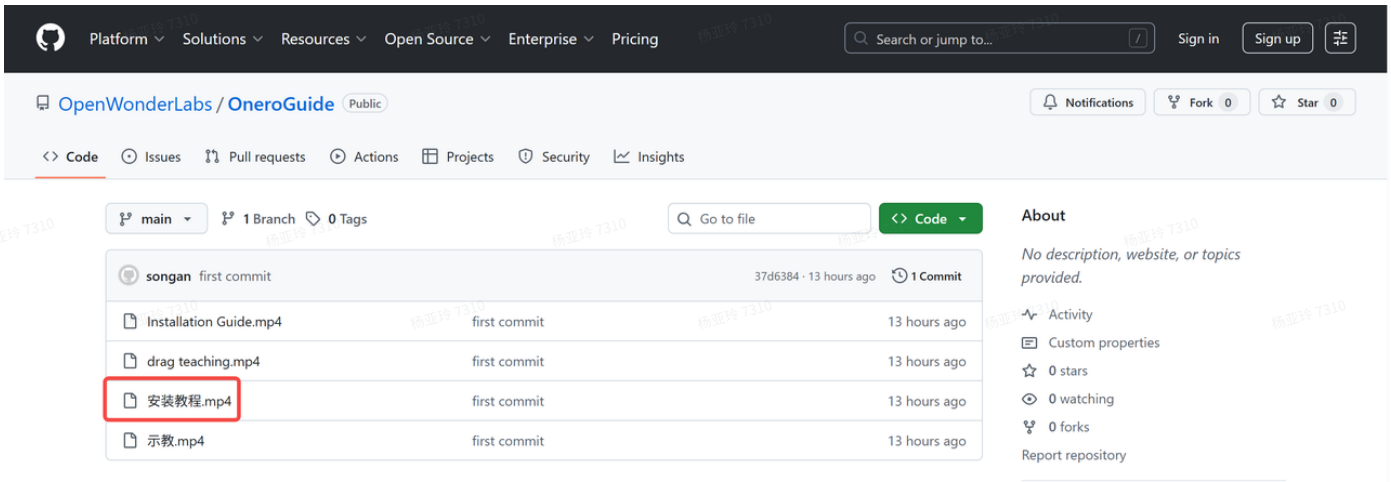
- English version: https://github.com/OpenWonderLabs/OneroGuide



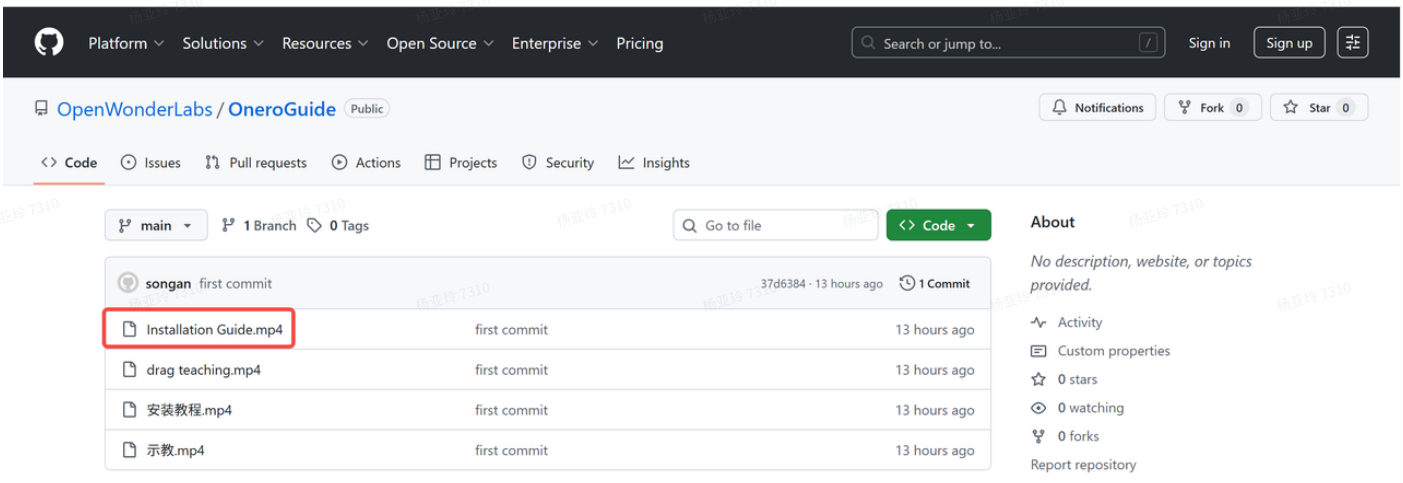
2.4 Basic Installation Guide

Please click the links below to watch the installation videos:

- Chinese version: https://github.com/OpenWonderLabs/OneroGuide



- English version: https://github.com/OpenWonderLabs/OneroGuide



3. Custom Development

The A1 Robotic Arm currently supports custom development via the Python SDK and ROS2 driver package. For detailed custom development instructions, please refer to the GitHub links below.

- C/C++ and Python projects: https://github.com/OpenWonderLabs/ArmApi
- ROS2 projects: https://github.com/OpenWonderLabs/ArmRos

4. Technical Specifications

Category	Item	Specification
Structure	Degrees of Freedom	7
	Maximum Payload	3 kg
	Arm Weight	4.3 kg
	Repeatability	±0.1 mm
	Maximum Reach	739.45 mm
	Rated Voltage	DC 48 V
	Rated Power Consumption	< 300 W
	Materials	Aluminum alloy frame, plastic housing
	Control Methods	Zero-force drag teaching / Offline trajecto
	External Interfaces	Power connector ×1, CAN connector ×1
	Mounting	M4 × 10 fixing screws
	Operating Environment	Temperature: 5 °C to 50 °C; RH 35–80% (n condensing)
	Noise Level	< 60 dB
Motion	J1 Range of Motion	−60° to +180°
	J2 Range of Motion	−180° to +15°
	J3 Range of Motion	±158°
	J4 Range of Motion	±110°
	J5 Range of Motion	±158°
	J6 Range of Motion	±90°
	J7 Range of Motion	±158°
	Maximum Joint Speed (J1–J7)	180°/s
Note: The above data are tested under controlled laboratory conditions. Actual performance may vary depend different environments and usage scenarios.		